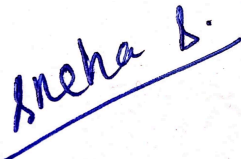
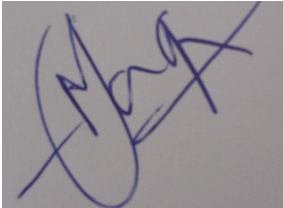


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3. Title of the invention:

AI Based Carbon Aware and SLA Aware Resource Scheduling

4. Technical field of the invention:

The current innovation belongs to cloud computing, artificial intelligence, and energy efficient computing systems. Particularly, the current invention is aimed at an intelligent scheduling of resources within the cloud data centers through the application of AI methods to optimize the performance at the lowest carbon footprint and the SLAs fulfilled.

5. Prior art:

The existing approaches to scheduling the cloud computing resources are primarily motivated by the aim of enhancing performance, reducing costs, and utilizing resources better. Nevertheless, all these strategies tend to ignore the impact scheduling decisions have on the environment. The environmental consideration is not a primary consideration in the cloud resource allocation/management process despite the fact that depending on the type of cloud computing, the energy usage and emissions may be immense. It is increasingly becoming an understanding that sustainability must feature in the formation of such scheduling techniques and most of the current methods exclude any form of carbon footprint reduction or energy saving measures.

This is an indication that there is a significant opportunity to reconsider resource scheduling, implying that the opportunity is not only technically and economically efficient but also contributes to more environmentally friendly practices. This may be solved by incorporating environmental issues with the traditional optimization criteria to create a more balanced approach on efficiency of a system and ecology. To conclude, although the emphasis has thus far been on performance and cost, it is a field that should be subject to more focus and creativity to include that of the environmental impact in the equation.

6. Objective

The main objectives of the present invention are:

- To develop an intelligent cloud resource scheduling system that reduces carbon emissions during task execution.
- To ensure that Service Level Agreement (SLA) requirements such as latency, response time and deadlines are strictly maintained.
- To integrate real time carbon intensity data and renewable energy availability into scheduling decisions.
- To improve resource utilization and reduce unnecessary energy consumption in cloud data centers.
- To design an adaptive system using artificial intelligence that continuously improves scheduling decisions using feedback.

7. Synopsis:

The proposed invention is simply a smart cloud resource scheduling system that attempts to make a reduction to carbon emission without violating SLA requirements. It resembles a structure with various modules interrelated with each other and assists in making quality decisions with regard to scheduling. The primary elements in this system are input layer, data collection unit, preprocessing unit, Artificial Intelligence based prediction and optimization unit, resource allocation unit and monitoring with feedback loop.

On starting, the system receives input such as user workloads, SLA constraints and cloud resources available. Such workloads may be in the form of AI training jobs, data processing jobs or generic applications. Then the system collects data on carbon intensity and the data on the availability of renewable energy provided by external sources. This section is quite significant since it assists the system to know when one energy is cleaner or not.

Then the second stage is preprocessing phase where raw data is washed and ready. Prediction module The role of machine learning to predict workload demand and carbon intensity is attempted. On the basis of this, optimization module determines the optimal procedure of allocating tasks based on balancing between carbon emission and SLA performance. Not only does it concentrate on the performance but also attempts to minimise energy wastage.

The resource allocation module allocates workloads to virtual machines and various servers of cloud infrastructure after a decision making process. It also supports dynamic VM allocation i.e. resources can be either increased or decreased as needed.

Distribution of work is done in such a way that distribution is done in areas or time of less impact of carbon.

Lastly, performance of SLA, energy use and carbon emission is monitored. This information is re-sent into the system so as to learn and make improved decisions in future. In general this invention resembles an intelligent scheduling system incorporating AI, real time carbon data and cloud management to make the computing environment more sustainable.

8. Brief description of drawings:

100 denotes user workload

This represents all the task requests coming from users or applications that need to be processed in the cloud system. These workloads can include data processing jobs, AI model training tasks, analytics operations or general computing tasks. Each workload has different requirements in terms of CPU, memory and execution time, which makes scheduling more complex.

101 denotes SLA requirement

This defines the performance limits that the system must follow while executing tasks. It includes parameters such as latency, response time and deadlines. The scheduler must ensure that all tasks are completed within these limits, otherwise SLA violation may occur which can affect service quality.

102 denotes cloud resource information

This includes the current status of available resources in the cloud environment such as CPU capacity, memory availability and virtual machine status. This information helps the system understand which resources are free, overloaded or underutilized before assigning tasks.

103 denotes data collection module

This module is responsible for gathering all required data from different sources. It acts as a central point where both system level data and environmental data are collected before further processing.

104 denotes carbon data source

This provides real time carbon intensity values and renewable energy availability data. It helps the system understand how much carbon emission is associated with energy usage at a given time, so that scheduling decisions can be made in a more eco-friendly way.

105 denotes cloud monitoring system

This module tracks resource utilization, energy usage and status of virtual machines continuously. It helps in identifying whether servers are underutilized or overloaded and provides important data for decision making.

106 denotes data processing and preprocessing

In this stage, raw collected data is cleaned, filtered and prepared before further use. Since data comes from multiple sources, it may contain inconsistencies or missing values, so preprocessing ensures the data is reliable and structured.

107 denotes prediction module

This module uses machine learning models to estimate future workload demand and carbon

intensity. It helps the system to predict how many tasks may arrive and what the carbon levels will be in future time slots, allowing proactive scheduling.

108 denotes optimization module

This module decides the best scheduling strategy by balancing carbon emission reduction and SLA requirements. It uses multi-objective optimization techniques to find the most efficient allocation plan.

109 denotes resource allocation

This module assigns tasks to suitable servers and virtual machines based on the decisions made by the optimization module. It ensures that resources are used efficiently while meeting performance constraints.

110 denotes dynamic VM allocation

This adjusts the number of virtual machines based on workload demand. If workload increases, more VMs are created, and if demand decreases, unused VMs are removed to save energy.

111 denotes feedback and monitoring

This module continuously checks system performance, energy usage and carbon emission during execution. It also provides feedback to the system so that future decisions can be improved.

112 denotes cloud infrastructure

This is the actual environment where all tasks are executed. It includes physical servers, storage systems and networking components which handle the workload processing.

113 denotes output

This represents the final results of the system such as optimized resource allocation, reduced carbon emissions and successful SLA compliance.

System workflow:

The system starts by receiving inputs in the form of user workload, SLA requirements and cloud resource information. These inputs define what tasks need to be executed, what performance limits must be followed and what resources are available.

After receiving inputs, the data is sent to the data collection module. Here, the system gathers environmental data such as carbon intensity and renewable energy availability along with system data like resource utilization and VM status.

Once the data is collected, it moves to the data processing and preprocessing stage where it is cleaned and structured properly. This ensures that the system works with accurate and meaningful data.

The processed data is then passed to the prediction module. This module predicts future workload demand and carbon intensity levels using machine learning models. These predictions help the system to plan ahead rather than reacting at the last moment.

Next, the optimization module uses this information to determine the best scheduling strategy. It balances between reducing carbon emissions and maintaining SLA performance. Based on this, it decides when and where tasks should be executed.

After optimization, the resource allocation module assigns tasks to appropriate servers and virtual machines. Dynamic VM allocation is also applied here, where the number of virtual machines is adjusted based on current workload demand.

Once tasks are allocated, they are executed in the cloud infrastructure. During execution, the feedback and monitoring module continuously tracks system performance, energy usage and carbon emissions.

The feedback collected is sent back into the system, creating a continuous learning loop. This helps the system improve its future scheduling decisions over time.

Finally, the system produces output in the form of optimized resource allocation, reduced carbon emissions and maintained SLA performance, showing that the system successfully balances efficiency and sustainability.

9. Detail description of the invention:

The present invention describes an AI based carbon aware and SLA aware resource scheduling system designed for modern cloud computing environments. The system is structured in a layered architecture where each module performs a specific function and all modules are connected in a continuous flow of data and decision making. The working of the invention begins with the input layer where three primary inputs are received, namely user workload, SLA requirements and cloud resource information. The user workload consists of task requests generated by users or applications which may include AI training jobs, data processing tasks or general computing operations. The SLA requirements define the expected performance limits such as latency, response time and execution deadlines. Cloud resource information provides the current status of available CPU, memory and virtual machine availability.

After receiving inputs, the system forwards this data to the data collection module. This module plays a very important role because it collects both system level and environmental data. The carbon data source gathers real time carbon intensity values from external APIs and also checks renewable energy availability. At the same time, the cloud monitoring system tracks resource utilization, energy consumption and the operational state of virtual machines. This combination of environmental and system data helps the system to understand both internal and external conditions before making decisions.

The collected data is then passed to the data processing and preprocessing module. In this stage, raw data is cleaned, filtered and structured properly because unprocessed data may lead to poor decision making. After preprocessing, the data is forwarded to the prediction module. The prediction module uses artificial intelligence techniques such as machine learning models to estimate future workload demand and carbon intensity patterns. This allows the system to predict when the workload demand will increase and when the carbon intensity will be high or low.

Based on these predictions, the system becomes proactive instead of reactive.

Next, the optimization module takes these predictions and performs multi objective optimization. The main aim here is to balance two important factors which are minimizing carbon emissions and maintaining SLA performance. The system identifies special scheduling opportunities and selects the special feasible option that satisfies each environmental and performance requirements. This step may be very important because it ensures that sustainability is carried out without impacting carrier high quality. After optimization, the system moves to the resource allocation module. In this stage, tasks are assigned to suitable servers and virtual machines. The system follows an energy efficient allocation strategy where low power servers are utilized first and high power servers are used only when required. The dynamic VM allocation mechanism allows the system to scale resources up or down depending on workload demand.

The allocated tasks are then executed in the cloud infrastructure layer which represents the real computing environment consisting of servers, storage and networking components. During execution, the feedback and monitoring module continuously tracks SLA performance, energy usage and carbon emissions. The feedback collected is again sent back to the system which creates a closed loop learning process. This enables the system to learn and improve its future scheduling decisions based on the outcome. As time passes, the system is able to operate in a more efficient, adaptive, and environmentally conscious way.

Finally, the output module displays the results, which can be the optimal resource usage, reduction in carbon footprint, and the maintenance of SLA.. This shows that the invention successfully balances performance and sustainability in cloud environments.

10. Best method of performance of the invention:

In real working, system is firstly received with the task requests and SLA requirements and resource availability information. It then gathers data on carbon intensity and renewable availability of energy externally. Then preprocessing is conducted to clean-up and organize data. Prediction module predicts the future workload needs and the carbon intensity level. On this information, optimization module determines the optimum scheduling policy using both the performance and environmental impact. There are those activities that can be postponed a bit to be done on the low carbon times as long as SLA permits. Then, resource allocation module allocates tasks to appropriate servers and virtual machines. Smaller and power efficient servers are deployed initially and power servers are deployed only when they are needed. This assists in minimizing unjustifiable usage of energy. Monitoring module is continuously checking compliance of SLAs and system performance during execution. In case any problem is identified, system will be able to correct resource allocation. Feedback loop assists system to learn the lessons of the previous choices and ameliorate in the long run. This is a way of efficient, reliable and eco friendly scheduling

11. CLAIMS:

1. A cloud resource scheduling system configured to minimize carbon emissions while maintaining SLA performance requirements.
2. The system as claimed in claim 1, wherein real time carbon intensity data and renewable energy availability are used for scheduling decisions.
3. The system as claimed in claim 1, wherein workload demand and carbon intensity are predicted using artificial intelligence models.
4. The system as claimed in claim 1, wherein a multi objective optimization module balances carbon emission reduction and SLA constraints.
5. The system as claimed in claim 1, wherein dynamic virtual machine allocation is used to improve energy efficiency and resource utilization.
6. The system as claimed in claim 1, wherein tasks are placed based on energy efficient workload placement strategies.
7. The system as claimed in claim 1, wherein a feedback and monitoring module continuously tracks SLA performance, energy consumption and carbon emission.

12. Inventive step of your invention:

The invention introduces a more advanced approach to cloud resource scheduling by combining artificial intelligence with carbon awareness and SLA constraints in a single unified system. Unlike existing scheduling techniques which mainly focus on performance or cost optimization, this system also considers environmental impact which is often ignored in traditional methods and leads to higher carbon emissions.

A key innovation in this invention is the use of a Deep Q Network based reinforcement learning model for decision making. The system is trained to analyze choicest scheduling strategies by means of observing workload patterns, carbon depth variations and system performance over the years. In preference to the use of fixed policies, the version constantly improves its decisions primarily based on revel in, which makes it greater adaptive and green.

In this invention, real time carbon intensity data is directly integrated into the scheduling process which allows the system to take environmentally informed decisions. The use of prediction models enables the system to forecast workload demand and carbon patterns in advance, which helps in better planning of resource usage and improves scheduling accuracy.

Another important innovation is the use of multi objective optimization where both carbon emission and SLA requirements are considered at the same time. This is different from traditional systems where only one objective is usually optimized. The system evaluates different scheduling possibilities and selects best balance between environmental impact and performance constraints.

To validate effectiveness of the proposed approach, the DQN based scheduling method is compared with several traditional scheduling algorithms such as First Come First Serve (FCFS), Round Robin, Greedy (least loaded), as well as more advanced approaches like LACS, CarbonClipper, and Temporal-Aware Carbon Smoothing Scheduler. These traditional and recent

methods generally operate based on static or predefined logic or limited optimization strategies and do not fully consider carbon intensity variations along with long term system impact.

The comparison is carried out based on important evaluation metrics such as total carbon emissions, energy consumption and SLA violation rate. The results show that the DQN based approach performs better in all these aspects. It produces lower carbon emissions by intelligently scheduling workloads during periods of lower carbon intensity and by optimizing resource utilization.

Another important advantage is that the proposed system reduces SLA violations by making more informed scheduling decisions. Unlike traditional algorithms which may delay tasks or overload certain servers, the DQN model learns to balance workload distribution in a way that respects SLA deadlines while still minimizing energy usage.

The invention also improves energy efficiency through dynamic virtual machine allocation and intelligent workload placement. The system prefers using low power resources first and activates high power servers only when required. This avoids unnecessary energy consumption and reduces idle power wastage which is common in existing methods.

Moreover, the comments primarily based studying mechanism lets in the system to continuously adapt and enhance over the years. Each scheduling selection contributes to destiny studying, making the device greater correct and stable with usage.

Because of these blended functions, the invention offers a clear technical advantage over current scheduling techniques by means of accomplishing lower carbon emissions, higher SLA compliance and stepped forward useful resource utilization. It additionally contributes to environmental sustainability through decreasing the general carbon footprint of cloud facts centers.

13. Industrial application:

The invention is applicable in large scale computing systems and AI based platforms as well as cloud data centers. It is applicable in the case of cloud services providers wishing to save power and minimize carbon emission without compromising service quality. It is also applicable in other industry sectors such as the finance, health, e commerce and real time data analytics whereby cloud systems are extensively deployed with performance mandates being crucial.

14. Abstract:

This invention introduces a carbon aware and SLA aware resource scheduling infrastructure to be deployed on the cloud. Carbon intensity real time data is collected by the system, predictive workload demand is made and optimization methods are employed to allocate resources. It minimizes carbon emissions and energy as well as preserving the performance of SLA. The system offers a dynamic and viable solution to the contemporary cloud computing environment.

15. Drawing

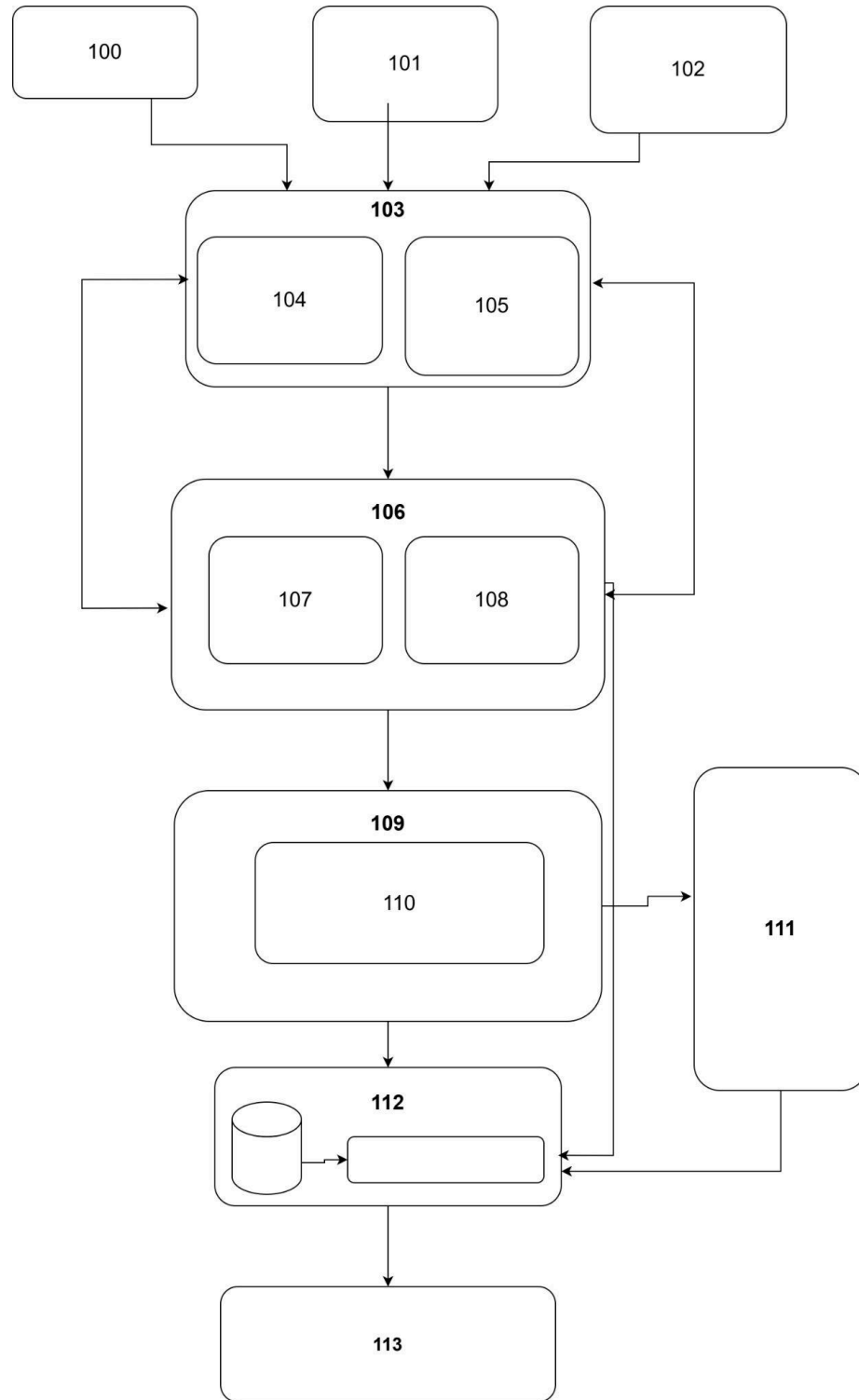


Figure 1 illustrates the overall architecture of the proposed carbon aware and SLA aware resource scheduling system.